Learning Objective 6.3: I can describe how HD Radio works and demodulate an HD Radio signal.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). From the lesson: a hybrid station's sidebands run from 129.4 to 198.4 kHz on each side of its center, subcarriers 356 to 546, spaced 363.4 Hz apart.

1. **Reading the display.** You tune `L14-hd-find.grc` to a station at 98.5 MHz.

   (a) Between which two frequencies, in MHz, should the upper sideband appear on the frequency sink? The lower sideband?

   upper

   lower

   (b) On the waterfall, the station at 98.5 MHz shows a rounded band whose width changes from moment to moment, and two straight bands of constant width on either side. Which is the analog host, and which are the HD Radio sidebands? What makes each look the way it does?

   (c) A second station shows shelves that start much closer to its host, about 102 kHz out, and end at 198.4 kHz. A third shows a rounded host and nothing beside it. Name the mode of each.

2. **Setting the OFDM blocks.** You adapt `L14-hd-test-tx.grc`, which runs at 1,488,375 samples per second with `fft_len` = 4096 and `cp_len` = 224, to a different sample rate.

   (a) At 2,976,750 samples per second, what `fft_len` keeps each FFT bin one 363.4 Hz subcarrier, and what `cp_len` keeps the guard the same length in time?

   `fft_len` ____

   `cp_len` ____

   (b) The OFDM Carrier Allocator's occupied carriers for the upper sideband are `list(range(356, 547))`. To build an extended hybrid station, which number do you change, in which direction, and what changes on the frequency sink?

   (c) You set the OFDM Cyclic Prefixer's rolloff from 112 to 0. What do you expect to see at the edges of the shelves, and why does it matter to a neighboring station?

3. **Setting the meter.** The sideband meter in `L14-hd-find.grc` uses a Frequency Xlating FFT Filter, Complex to Mag^2, a Moving Average of `avg_len` samples (0.2 s), Keep 1 in N, and Log10 for each band.

   (a) You shorten the Moving Average to 0.02 s. What changes in the readings, and what stays the same?

   (b) You type `sb_center` into the lower sideband's filter instead of `-sb_center`. What do the two readings show?

   (c) On an extended hybrid station, you leave `inner` at 356. Does the meter read high or low, and what setting fixes it?

4. **What noise does.** In `L14-hd-test-tx.grc` you raise the Variable `noise_amp`, run it again, and play the new file in the sideband finder.

   (a) Describe what happens to the floor, the shelves, and the host on the frequency sink as `noise_amp` goes up.

   (b) What happens to the two meter readings, and why?

   (c) On a real station, what would you expect nrsc5, and a car radio, to do as the shelves sink into the floor?